Viet Nguyen

3+ years of experience GIS Analyst

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Personal website
Vietnamese
30.05.1997 (He/ Him/ His)
LinkedIn
GitHub

Work experience

11/2023 - present
Greifswald, Germany

GIS Analyst
Institute of Geography and Geology, University of Greifswald

- Analyzed spatial data with Python (GeoPandas, NumPy, GDAL, Rasterio, Multiprocessing, etc.), and R (sf, terra, lidR, rgdal, doParallel, etc.)
- Performed drone-based data acquisition (spectral, RGB, and LiDAR data)
- Processed drone-based data with Agisoft Metashape
- Developed WebGIS applications using HTML, CSS, and JavaScript (Leaflet)

01/2021 - 08/2023
Eberswalde, Germany

GIS Technician *(TRANSECT,* Virtual Forests *project)*
Centre for Econics and Ecosystem Management, Eberswalde University

- Implemented Random Forest algorithm using Python with Landsat 8, Sentinel 1, and Sentinel 2 data in Google Earth Engine. Achieved 91% classification overall accuracy
- Collected and digitized on-screen training and validation data for landcover classification using QGIS
- Interpreted multi-temporal remote sensing metrics (NDVI, NRPB, VV, VH, etc.)
- Produced maps, figures, and tables using R and QGIS
- Analyzed statistics with R (dplyr, ggplot2)

10/2021 - 05/2023
Eberswalde, Germany

Intern in
Forest inventory based on ALS point cloud on large-scale levels *(WINMOL project)*
Thünen Institute of Forest Ecosystems

- Developed methods to derive individual tree attributes (coordinate, tree height, diameter at breast height (DBH), crown base height, crown area) from ALS point clouds using R (lidR, rLiDAR, TreeLS, etc.), achieved 78% tree detection rate
- Evaluated statistical models for tree attributes using R, attained 0.86 R^2 for tree height estimates, and 0.74 R^2 for DBH estimates

Education

10/2020 - 06/2023
Eberswalde, Germany

M.Sc. in Forest Information Technology
Eberswalde University for Sustainable Development

- GPA: 1.0 (German grade)

09/2015 - 06/2019 Hanoi, Vietnam	**B.Sc.** in Soil Science Vietnam National University

Professional training

2024	***Structure-from-Motion photogrammetry,*** The University Centre in Svalbard
2024	***EnMAP data access and image preprocessing techniques***, EO College
2024	***Introduction to Hyperspectral Remote Sensing***, EO College
2023	***Automating GIS Processes,*** University of Helsinki
2023	***Geo-Python,*** University of Helsinki
2023	***Elements of AI,*** University of Helsinki
2021	***Getting started with Python,*** University of Michigan
2021	***Using Python to Access Web Data***, University of Michigan
2021	***Python Data Structures***, University of Michigan
2021	***Introduction to R,*** DataCamp

Skills and License

GIS	QGIS, ArcGIS, Google Earth Engine
Programming	R, Python
WebGIS	HTML, CSS, JavaScript (Leaflet)
Database	SQL, PostgreSQL, PostGIS
Digital skills	Microsoft Office, Git, GitHub, Data Management
License	EU Drone License A1-A3

Other skills

Soft skills	Team spirit and independent working, growth mindset, cultural awareness, reliability, and ability to work under pressure
Languages	Vietnamese (Native), English (C2), German (A2)

November 2024

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